

Amber Wang

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EDUCATION

University of Pennsylvania

MSE in Systems Engineering

Aug 2024 – May 2026

University of California, San Diego

B.A. in Economics

Sep 2020 – Jun 2024

PROFESSIONAL EXPERIENCE

Business Analyst | Penn Biotech Group | Philadelphia, PA

Sep 2025 – May 2026

- Assessed manufacturing workflow, material utilization and cost data using Excel, uncovering 15–20% process inefficiencies to support production planning, capacity analysis and operational expansion decisions for medical device manufacturing.
- Built unit economics and cost-modeling dashboards in Excel, performing scenario analysis on material, labor and overhead assumptions to estimate production profitability and improve gross margin by up to 12%.
- Evaluated 10+ suppliers based on cost, technical capabilities, quality considerations and production readiness, prioritizing 5 partners to support sourcing decisions and manufacturing scale-up.

Supply Chain Business Analyst Intern | QPO Energy | Tualatin, OR

May 2025 – Jul 2025

- Developed Excel-based BOM, inventory and cash flow models (PivotTables, INDEX/MATCH, scenario analysis) to evaluate material demand, procurement costs and inventory risks, enhancing forecasting accuracy and enabling earlier detection of shortages by 23%.
- Reviewed purchasing costs and supplier lead-time data to identify bottlenecks, optimize inventory planning and reorder strategies to reduce supply delays by 15% through data-driven reorder point and safety stock analysis.
- Automated order tracking and reporting workflows with VBA and rule-based checks, improving data accuracy and reducing processing time by 55% to streamline procurement, operations and finance decision-making.

Business Analyst | Penn Graduate Consulting Club | Philadelphia, PA

Sep 2024 – May 2025

- Synthesized and visualized pilot study data and user feedback using Excel and Power BI to surface usability trends and prioritize product enhancements, contributing to a 34% increase in user satisfaction.
- Conducted user behavior, adoption and survey analysis to measure feature effectiveness, revealing the top 3 drivers of user engagement, informing product enhancement priorities and supporting data-driven decision-making.
- Benchmarked competitor products and market positioning, building structured comparison dashboards and synthesizing insights into recommendations that informed strategic planning, resource allocation and program improvements.

Student Financial Analyst | UCSD Costing Policy & Analysis | San Diego, CA

Sep 2022 – Jun 2024

- Designed Excel-based financial and operational models to support internal service rate planning, resource allocation and budgeting decisions across \$60M+ in annual expenditures, increasing reporting consistency of 30+ weekly internal transactions.
- Coordinated with 50+ campus units to validate financial and operational data using standardized reconciliation templates, improving data accuracy and supporting cross-functional reporting and resource planning.
- Standardized reporting templates, documentation and data handoff processes across departments, reducing manual errors by 20% and improving data integrity, reporting consistency and cross-functional visibility.

SELECTED PROJECTS

Game Pricing & Demand Analytics | Product & Pricing Analytics

Sep 2025 – Dec 2025

- Analyzed 200K+ Steam applications and 1M+ user reviews using Python, conducting data cleaning, exploratory analysis and trend analysis to uncover key demand drivers across product attributes, genres and customer engagement metrics.
- Developed predictive analytics and forecasting solutions to evaluate pricing opportunities and demand patterns, generating insights on user behavior and price sensitivity to inform pricing strategy, product positioning and planning decisions.

Readmission Risk Modeling | Healthcare Operations Analytics

Sep 2025

- Investigated 9,000+ EHR records using Python to identify patterns, trends and key risk factors, supporting data-driven planning and resource allocation decisions while improving visibility into high-risk patient populations.
- Created an operational decision-support tool that integrated risk assessments into existing workflows, providing real-time visibility into high-risk cases to support workforce planning and case prioritization.

SKILLS

- Supply Chain & Planning:** Demand Forecasting, Inventory Planning, Inventory Optimization, Supply Planning, Material Requirements Planning, Production Planning, BOM Management, Procurement Analytics, Lead Time Analysis, Safety Stock Analysis, KPI Tracking
- Analytics & Methods:** Forecasting, Scenario Analysis, Statistical Analysis, Exploratory Data Analysis, Cost Analysis, Inventory Analysis, Root Cause Analysis, Dashboard Development, Data Cleaning & Validation, Process Improvement, Workflow Optimization
- Data & Technical Tools:** SQL (MySQL, PostgreSQL), Python (Pandas, NumPy), Power BI, Tableau, Advanced Excel (PivotTables, XLOOKUP/VLOOKUP, VBA, Power Query), SharePoint, Microsoft Office Suite